SinkAware: Approximate Low-Rank Fixed-Sink Priors for Attention

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Abstract

Attention sinks are fixed early tokens that often receive substantial attention. A tempting systems idea is to precompute or reuse their contribution. We show that exact static reuse is invalid under standard softmax attention because fixed sink-token logits remain query-dependent. We therefore test a narrower approximation: per-head low-rank predictors for fixed sink logits while keeping all non-sink scores exact. On Mac-local distilgpt2 traces, rank-2 prediction reduces aggregate held-out attention-output relative L2 error to 0.141 versus 0.170 for a position-only predictor, while the cost model estimates 0.531x exact four-sink QK multiply-adds. Layer-head and head-selection analyses weaken broad claims, but split/seed, length, trace-level, and GPT2/OPT-family gates keep the all-head rank-2 row positive. The strongest Mac-local stability gate keeps all four model/length rows positive at lengths 64 and 96, with mean output relative-L2 improvement 0.0535 ± 0.0262 and minimum row improvement 0.0301. The approximate operator passes Triton interpreter correctness tests against its CPU reference. A first downstream causal-LM control smoke on distilgpt2 and OPT-125M also passes: exact replacement is a no-op, and rank-2 is closer than position-only in loss drift and KL on both models, with aggregate absolute loss-delta improvement 0.1008 ± 0.1166. This is not cross-model predictor transfer, benchmark success, or a GPU speed result. It is a bounded correctness, stability, and downstream-control gate for deciding whether approximate sink prediction is worth native implementation.

1 Problem

For query q_t, sink keys K_s, and non-sink keys K_n, the sink logits $q_t K_s^\top$ are still query-dependent. Any exact method that skips them changes the softmax denominator or attention weights. SinkAware kills that exact static-prior branch and revives only an approximate one.

2 Approximate Branch

The live branch predicts fixed sink-token logits from a low-dimensional query feature. Non-sink logits remain exact. The branch is useful only if the approximation is cheap enough and attention-output drift stays bounded. Table 1 reports the original aggregate layer means. This view keeps rank-2 alive: it improves output relative L2 and sink-mass error over position-only while staying under the simple multiply-add estimate for exact four-sink QK.

Table 2 is the reviewer-facing caveat. The same probe now reports layer-head paired improvements over the position-only predictor. Positive values mean lower error than position-only. Rank-2's mean output improvement is positive, but the interval overlaps zero and the win rate is only 0.278. This makes the branch weakly alive, not established.

We also tested the obvious Mac-local rescue: select rank-2 only for heads where rank-2 beats position-only on a validation split, then evaluate that mixed policy on held-out tokens. The rule selected 19/72 heads but failed held-out: output relative L2 was 0.204 for

Predictor	Sink RMSE	Sink-mass MAE	Attention L1	Output rel-L2
static	1229.509	0.095	0.217	0.193
position	1197.390	0.076	0.175	0.170
rank1	1079.541	0.067	0.157	0.160
rank2	1001.506	0.055	0.135	0.141
rank4	838.320	0.045	0.115	0.122
rank8	699.232	0.040	0.104	0.107

Table 1: 48-trace distilgpt2 softmax/output probe. Lower is better.

Predictor	Output rel-L2 improve	95% CI	Win rate	Sink-mass improve
rank1	0.0257	0.0290	0.250	0.0086
rank2	0.0297	0.0378	0.278	0.0206
rank4	0.0596	0.0465	0.333	0.0308
rank8	0.0955	0.0805	0.347	0.0360

Table 2: Layer-head paired improvements over position-only across 72 layer-head cells. Positive is better; intervals are normal 95% intervals over layer-head cells.

validation-selected rank-2, compared with 0.172 for position-only and 0.142 for all-rank2. This rules out simple validation head selection as a pre-GPU rescue.

The next stability gate repeated all-head rank-2 under randomized token train/test splits. Across seeds 0, 1, and 2, rank-2 beat position-only on every split, with mean output relative-L2 improvement 0.0368 ± 0.0006, sink-mass MAE improvement 0.0203 ± 0.0003, and attention-L1 improvement 0.0435 ± 0.0004. This supports the all-head aggregate row enough for interpreter work, but not a per-head robustness claim: the output win rate across layer-head cells remains 0.282 ± 0.024.

We then ran a bounded sequence-length and sink-count sweep. For max lengths 64 and 96, sink counts 2 and 4, and three token split seeds per configuration, all configurations kept rank-2 positive against position-only. Mean output relative-L2 improvement across configurations was 0.0366 ± 0.0024, and the minimum configuration improvement was 0.0342. This makes the all-head row more repeatable, but the per-head caveat persists: layer-head output win rate averaged only 0.286 ± 0.010.

Finally, we ran a larger trace-level frozen split repeat. Across three deterministic splits of 48 traces, each split trained on 32 whole traces and evaluated on 16 held-out traces. Rank-2 beat position-only on every split: output relative-L2 improvement averaged 0.0379 ± 0.0014, sink-mass MAE improvement averaged 0.0220 ± 0.0008, attention-L1 improvement averaged 0.0461 ± 0.0014, and the minimum output improvement was 0.0367. This is a stronger repeatability check than token-level splitting, but it does not remove the per-head caveat because the layer-head output win rate remains only 0.278 ± 0.016.

Before local Triton execution was unblocked, we ran a larger held-out/model-family falsification repeat. The gate fits all-head rank-2 predictors separately per model, then compares rank-2 against position-only on held-out traces. On 48 traces and split seeds 0, 1, and 2, distilgpt2 improves by a measured 0.0306 ± 0.0023 output relative-L2, with minimum split improvement 0.0283. facebook/opt-125m improves by a measured 0.0788 ± 0.0069, with minimum split improvement 0.0731. The aggregate model-row improvement is 0.0547 ± 0.0472, with minimum model improvement 0.0306. This is not cross-model predictor transfer and is not promotion evidence, but it is a stronger falsification check than the prior one-seed smoke because the rank-2 row did not collapse under repeated OPT-family held-out splits.

We then broadened that same family gate across sequence length. With 48 traces, lengths 64 and 96, sink count 4, and split seeds 0, 1, and 2, every model/length row remained positive. The aggregate model/length output relative-L2 improvement was 0.0535 ± 0.0262, with minimum row improvement 0.0301 and layer-head output win rate 0.982 ± 0.008. The per-row improvements were 0.0306 ± 0.0023 and 0.0301 ± 0.0018 for distilgpt2 at lengths 64 and 96, and 0.0788 ± 0.0069 and 0.0744 ± 0.0040 for facebook/opt-125m at lengths 64 and 96. This strengthens the Mac-local stability decision surface, but it still does not measure downstream quality, predictor transfer, or GPU speed.

Downstream control smoke. The next Mac-feasible gate patches full GPT2/OPT attention during causal-LM evaluation. It compares exact baseline attention against exact sink-logit replacement, position-only replacement, and rank-2 replacement. Exact replacement is the validity control and reproduces baseline loss and logits on both tested models. Rank-2 is closer than position-only on both models: distilgpt2 loss drift falls from 0.0941 to 0.0527, and facebook/opt-125m loss drift falls from 0.2673 to 0.1070. Across the two rows, aggregate absolute loss-delta improvement is 0.1008 ± 0.1166, aggregate KL improvement is 0.0815 ± 0.1032, and the minimum model loss improvement is 0.0414. This keeps the branch alive beyond isolated attention-output drift, but only as a small downstream-control smoke: predictors are still fit separately per model, the slice is small, and no benchmark accuracy or GPU speed is measured.

3 Reference and Kernel Gates

The Phase 3 reference specifies the exact computation to preserve during kernel work. Given query q, keys K, values V, and predicted fixed-sink logits $\hat{s}$, SinkAware computes all tail logits $qK_{tail}^{\top}$ exactly, replaces only the first sink-token logits with $\hat{s}$, and then runs the ordinary softmax denominator over the concatenated vector. Unit tests verify that when $\hat{s}$ equals exact sink logits, the reference reproduces exact attention to numerical tolerance. This keeps the live branch separate from the killed exact-static-reuse branch.

The Phase 4 Triton-interpreter scaffold now targets the same operator for a one-head scalar-output synthetic gate. Triton's official debugging documentation describes interpreter mode as a CPU simulation path enabled by `TRITON_INTERPRET=1`, bypassing compilation for debugging.[1] The local Mac environment now uses a repo-local source build of the experimental `triton-cpu` backend, and the SinkAware Phase 4 interpreter tests pass. This closes the Mac kernel-logic correctness gate before native timing work, but it is not evidence of GPU performance.

4 Decision

SinkAware is alive only as an all-head approximate low-rank correctness and native-GPU gate. Exact static reuse remains dead, and the simple head-selective rescue is weakened. The larger trace-level frozen repeat strengthens the all-head row, the repeated OPT-family length gate avoids an immediate model-family or length collapse, the Triton interpreter gate passes, and the first downstream control smoke favors rank-2 over position-only. Promotion still requires larger downstream controls plus native runs that show real speed or memory-traffic advantage over exact attention while still beating the position-only predictor on quality.

5 Artifacts

- `experimental/sinkaware/phase2/real_qk_sink_softmax_output_probe.md`
- `experimental/sinkaware/phase2/head_selective_sink_gate.md`
- `experimental/sinkaware/phase2/rank2_split_stability_gate.md`
- `experimental/sinkaware/phase2/rank2_length_sink_sweep_gate.md`
- `experimental/sinkaware/phase2/rank2_trace_frozen_split_gate.md`
- `experimental/sinkaware/phase2/rank2_cross_model_falsification_gate.md`

[1] `https://triton-lang.org/main/programming-guide/chapter-3/debugging.html`

- experimental/sinkaware/phase2/rank2_cross_model_length_stability_gate.md
- experimental/sinkaware/phase2/qk_sink_cost_model.md
- experimental/sinkaware/phase3/approx_sink_attention_reference.md
- experimental/sinkaware/phase3/downstream_quality_control_gate.md
- experimental/sinkaware/phase4/approx_sink_attention_triton_gate.md
- experimental/sinkaware/phase2/gpu_gate_runbook.md
- experimental/triton_cpu_source_install_20260506.md
- experimental/sinkaware/paper/reviewer_pack.md